

The resting membrane potential

These notes follow the lecture slide by slide. Each section names the slides it covers, so you can read it before the video, follow it during the video, or use it afterward to check what you remember. The four worked problems from the slides are written out in full here. Silverthorn chapter 5 (the resting membrane potential) and chapter 9.

COMPETENCIES 1 TO 5 OF WEEK 4

26 SLIDES · 4 WORKED PROBLEMS · 1 MOVING MODEL

HOW TO USE THIS SHEET

Pass one, before the video: read the section, then close the page and put what you can into the box for that competency on your Competency Study Guide. Pass two, after the video: come back in a different color and add what you missed. Try each worked problem yourself before you read the work. Every one asks you to predict a direction and explain it, so a pencil is all you need.

The problem for today

SLIDES 1 TO 2 OF THE DECK

Every cell in your body has a voltage across its membrane. It is small, about 70 thousandths of a volt, and the inside of the cell is the negative side. In this lecture we build that voltage starting from zero. We start with a cell that has no voltage at all, add one membrane protein at a time, and read a voltmeter after each step, so that you can see what each protein contributes. Then we read two equations in plain words and use them to predict which way ions move, and last we change the blood chemistry and predict what happens.

Here is the problem we solve by the end. A construction worker is pinned under a collapsed wall for five hours with both legs crushed. He is awake and talking the whole time. The rescue team lifts the wall, and within minutes his heart rhythm turns chaotic and his heart stops. Nothing new injured him. The only thing that changed is that blood began flowing through his legs again.

Blood drawn in the ambulance shows a plasma potassium of 7.0 mM, where the normal range is 3.5 to 5.0 mM. His sodium (139 mM, range 135 to 145), chloride (102 mM, range 98 to 106), glucose (94 mg/dL, range 70 to 100) and temperature, 36.8 °C (98.2 °F), are all in range. So potassium rose by about 3 mM and everything else held. That does not sound like much, and sodium can move by 3 mM without anyone noticing. Write down your best guess for why 3 mM of potassium can stop a heart. By the end of these notes you will be able to explain the answer.

See it move: build the resting potential yourself

SLIDE 20 OF THE DECK

This is a working model of the cell in this lecture. Press the steps in order, 1 to 6, and watch the ions and the voltmeter. Each step matches a section of the notes below, so keep it open and come back to it as you read. Step 2 goes with the pump, step 3 with the potassium leak channels, step 4 with the sodium cell, step 5 with the real resting cell, and step 6 is a preview of the action potential. You can pause it at any time, turn the pump on and off, and change the potassium outside the cell with the slider.

Moving model: run it on the course site, [drsrennie-stack.github.io/human-physiology-Fa26/biol005-w04-membrane-potential-notes.html](https://stack.github.io/human-physiology-Fa26/biol005-w04-membrane-potential-notes.html)

THREE THINGS TO TRY

- 1 On step 3, wait for the voltmeter to settle near -95 mV and then watch the potassium channels. Ions are still crossing in both directions. That is what no net movement looks like.
- 2 On step 5, slide the potassium outside the cell from 4 up to 7 mM, and then down to 2.5 mM. Write down the voltmeter reading each time and compare it with worked problem 4.
- 3 On step 5, turn the pump off and watch for a full minute. The voltage barely moves at first and then drifts toward zero as the gradients run down. A real cell takes minutes to hours to do this.

Two terms, and what a voltmeter reads

SLIDES 3 TO 4 OF THE DECK

Two terms need to be kept apart from the start. The resting membrane potential is a steady voltage across the membrane of a cell that is not being stimulated. Almost every cell in the body has one: liver cells, fat cells and skin cells as well as neurons and muscle cells. An action potential is a brief, rapid change in that voltage that travels along the membrane. Only excitable cells can produce one, which means neurons and the three kinds of muscle. A liver cell has a resting potential because it has the sodium potassium pump and potassium leak channels, which are covered below. It cannot fire an action potential because it does not have voltage gated sodium channels. This lecture covers the resting potential. The action potential is the next lecture, and you need this one to understand it.

A membrane potential is always a comparison. One electrode goes inside the cell, one stays outside, and the meter reports the inside compared with the outside. By agreement the outside is called zero, so -70 mV means the inside is 70 millivolts more negative than the outside. A millivolt is one thousandth of a volt. A AA battery is 1,500 mV, so a cell holds about one twentieth of that across a membrane only 8 nanometers thick. Typical resting values are about -70 mV in a neuron and about -90 mV in skeletal and heart muscle.

One point is easy to get wrong. The cell is not filled with negative charge. The fluid inside and the fluid outside are each electrically neutral, with equal numbers of positive and negative charges. The only place the charges are unequal is a very thin layer on the two

faces of the membrane, slightly more negative charge on the inner face and slightly more positive charge on the outer face.

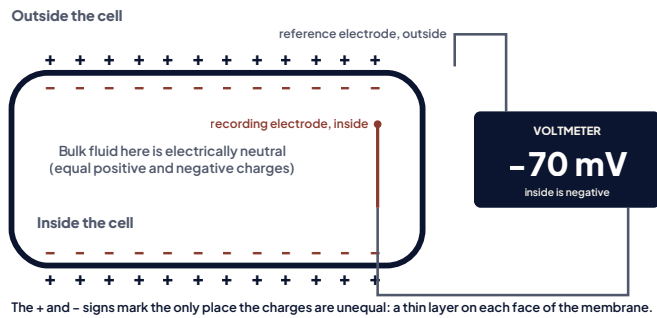


Figure 1. One electrode inside, one outside. The meter reports the inside compared with the outside.

Resting membrane potential

The steady voltage across the membrane of an unstimulated cell, inside compared with outside. About -70 mV in a neuron.

Action potential

A brief, rapid change in membrane potential that travels along the membrane. Only neurons and muscle cells produce one.

Millivolt (mV)

One thousandth of a volt.

Excitable cell

A cell that can fire action potentials: neurons and skeletal, cardiac and smooth muscle cells.

Where the membrane proteins come from

SLIDE 5 OF THE DECK

Every pump and channel in this lecture is a protein, and every protein is built from a gene. All of your cells contain the same genes. What differs from one cell type to another is which genes are switched on. All cells switch on the genes for the sodium potassium pump and for potassium leak channels, which is why almost every cell has a resting potential. Neurons and muscle cells also switch on the genes for voltage gated sodium and potassium channels, which is why only they can fire action potentials.

A membrane protein follows a fixed route from its gene to the membrane. In the nucleus the gene is copied into mRNA. Ribosomes on the rough endoplasmic reticulum read the mRNA and build the protein directly into the ER membrane. The Golgi finishes and sorts the protein. A vesicle carries it to the surface, and when the vesicle fuses with the cell membrane, the protein becomes part of that membrane. Transcription and translation are on the review page, so those steps are not repeated here.

A cell can change how many copies of a channel are in its membrane by adding more or taking some out. More open channels for an ion means a higher permeability to that ion, and permeability decides what the membrane potential will be. This is also why a mutation in one channel gene causes disease mainly in the tissues that use that gene. One inherited form of long QT syndrome comes from a faulty potassium channel in heart muscle: heart cells take longer to repolarize, and the person is at risk of dangerous rhythms. Diseases caused by a faulty ion channel gene are called channelopathies. Cystic fibrosis is another one, caused by a faulty chloride channel.

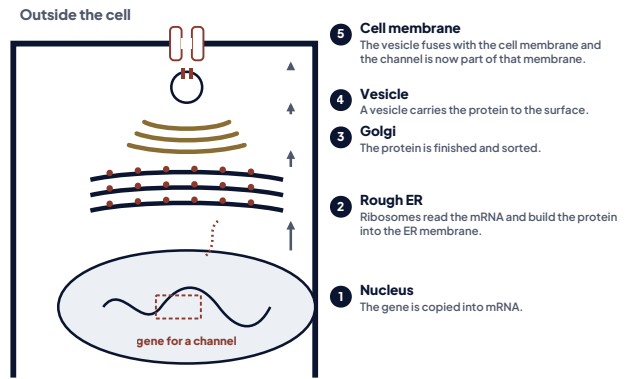


Figure 2. The route a channel or pump takes from its gene to the cell membrane.

Gene expression

Using a gene to build its protein. Cells differ in which genes they express, not in which genes they have.

Channelopathy

A disease caused by a faulty ion channel gene, such as some forms of long QT syndrome or cystic fibrosis.

Building the voltage: the pump

SLIDES 6 TO 8 OF THE DECK

Start with an imaginary cell that has a lipid membrane and nothing else. Potassium, sodium and everything else are mixed the same way inside and outside, and ions cannot cross the oily middle of a bilayer, so nothing moves. Inside there are large negative molecules that can never leave, proteins and the phosphate groups on ATP and nucleic acids, but each one is paired with a positive ion, so the inside is neutral. The voltmeter reads zero. For a voltage to appear, positive and negative charges have to end up unequally shared across the membrane. That takes two things: something to make the two sides different, and a way for some ions, and not others, to cross.

The first protein we add is the sodium potassium pump, the Na^+/K^+ ATPase. It is in the membrane of every cell you have. Each cycle moves 3 sodium ions out and 2 potassium ions in, and splits one ATP to pay for it, because both ions are being moved toward the side where they are already more concentrated. The pump uses a lot of energy: roughly a fifth to a third of the ATP a resting body uses, and more than that in neurons.

Because three positive charges leave for every two that enter, the pump does make the inside slightly negative. A pump that moves net charge like this is called electrogenic. But look at the voltmeter. Even with the pump running at full speed, the cell is only a few millivolts negative. A real neuron is at -70 mV , so the pump alone does not explain the resting potential.

The main thing the pump does is create two steep concentration gradients. Potassium ends up at about 140 mM inside and 4 mM outside, 35 times more concentrated inside. Sodium ends up at about 145 mM outside and 15 mM inside, roughly 10 times more concentrated outside. Chloride is about 110 mM outside and 10 mM inside, and free calcium is about 1 mM outside and only 0.0001 mM inside. The inside of a cell is high in potassium and the fluid around it is high in sodium. A concentration difference is a form of stored energy, like water held behind a dam, and the

pump uses ATP to build it. No ions cross yet, because there is still no channel for them to cross through.

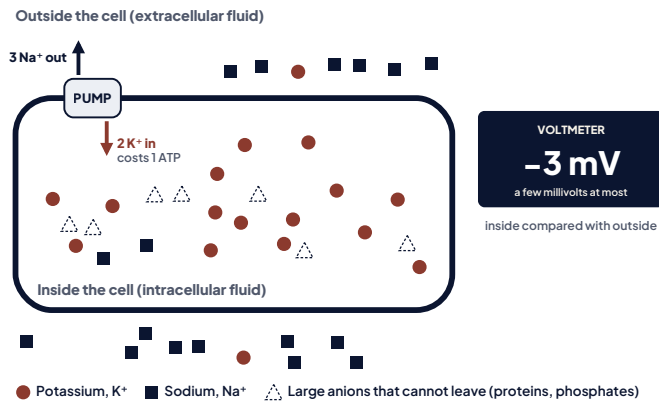


Figure 3. The pump sorts the ions and produces very little voltage on its own.

Sodium potassium pump (Na^+/K^+ ATPase)

A membrane protein that moves 3 Na^+ out and 2 K^+ in for each ATP it splits. It builds and maintains the ion gradients.

Electrogenic

Describes a pump that moves net charge across the membrane and so contributes directly to the membrane potential.

Concentration gradient

A difference in the concentration of a substance between two places. Substances diffuse from the higher concentration toward the lower.

Typical concentrations

K^+ 4 mM outside, 140 inside. Na^+ 145 outside, 15 inside. Cl^- 110 outside, 10 inside. Free Ca^{2+} about 1 outside, 0.0001 inside.

Potassium leak channels and the equilibrium potential

SLIDES 9 TO 11 OF THE DECK

The second protein we add is a potassium leak channel. A leak channel is a protein pore with no gate, so it is open all the time. This one is lined so that potassium fits through and sodium does not. The channel does not choose a direction. It only makes movement possible. Potassium is 35 times more concentrated inside, so it diffuses out.

Each potassium ion that leaves carries one positive charge out and leaves behind the negative partner it was paired with. Those large anions are too big for any channel. So every potassium ion that leaves makes the inner face of the membrane a little more negative. A voltage produced this way, by an ion diffusing down its concentration gradient, is called a diffusion potential.

Now two forces act on every potassium ion. The chemical force comes from the concentration gradient and pushes potassium out. The electrical force comes from the voltage: a negative interior attracts a positive ion and pulls potassium back in. Together they are the electrochemical gradient. At the start the electrical pull is very small, so potassium leaves quickly. Each ion that leaves makes the inside more negative, and that makes it harder for the next one to leave. At one particular voltage the two forces are exactly equal and opposite. Potassium ions still cross in both directions, but as many enter as leave, so there is no net movement, even though the channels are still open. That voltage

is the equilibrium potential for potassium, written E_K , and it is also called the Nernst potential. With 140 mM inside and 4 mM outside it is about -95 mV .

Many students picture potassium leaving until the cell is nearly empty. That is not what happens. Setting up -95 mV across the membrane of a typical cell takes far fewer than one potassium ion out of every thousand in the cell, and the concentrations do not measurably change. The reason so few ions can do so much is where they sit. The potassium ions that left stay pressed against the outer face of the membrane, held by the negative charges they left on the inner face only 8 nanometers away. Opposite charges held that close together across a thin insulator produce a large voltage from a small number of charges, which is how a capacitor works. This is why a membrane potential can change in a thousandth of a second, and why a cell can do it again and again without running down.

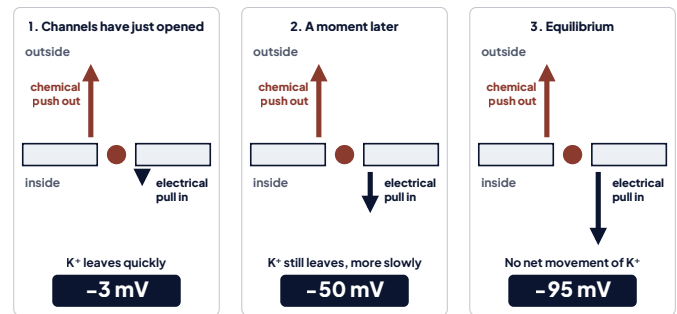


Figure 4. The push from concentration stays the same. The pull from voltage grows with every ion that leaves, until the two are equal.

Leak channel

An ion channel with no gate. It is always open. Potassium leak channels are the most numerous and set the resting potential.

Diffusion potential

A voltage produced across a membrane when an ion diffuses down its concentration gradient and leaves its oppositely charged partner behind.

Electrochemical gradient

The combined effect of the chemical force (concentration) and the electrical force (voltage) on an ion.

Equilibrium potential (Nernst potential)

The membrane potential at which the electrical force on an ion exactly balances its concentration gradient, so there is no net movement of that ion. E_K is about -95 mV .

A sodium cell, and one voltage scale

SLIDES 12 TO 13 OF THE DECK

Take a second imaginary cell. Give it the same pump, so it has the same gradients, but close the potassium channels and open channels that only sodium fits through. Sodium is concentrated outside, so the chemical force pushes it in. At the start the inside is slightly negative, so the electrical force pulls it in as well. Both forces point the same way, so sodium enters quickly. Each sodium ion brings a positive charge with it, so the inside goes from slightly negative, through zero, to positive. Once the inside is positive, the electrical force reverses and begins pushing sodium back out. The two forces balance at about $+61 \text{ mV}$, the equilibrium

potential for sodium, E_{Na} . The same rules gave the opposite sign, because the gradient points the opposite way.

Put the two cells on one voltage scale. Whenever an ion can cross the membrane, its movement shifts the membrane potential toward that ion's equilibrium potential: -95 mV for potassium and $+61$ mV for sodium. The more open channels an ion has, the larger its effect, so the ion with the highest permeability has the most influence on the membrane potential. A resting neuron sits at -70 mV, which is 25 mV from E_K and 131 mV from E_{Na} . That tells you the resting membrane is far more permeable to potassium than to sodium, roughly 25 to 40 times more. It is not exactly at E_K , which means a small amount of another ion is also crossing. That ion is sodium.

Where each ion would hold the voltage if it were the only ion that could cross

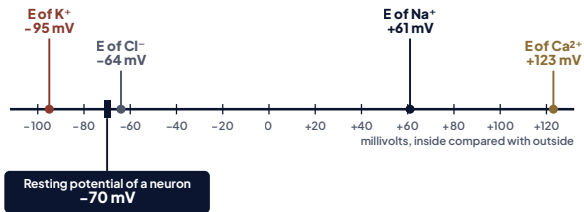


Figure 5. Equilibrium potentials for the typical concentrations, with a neuron's resting potential marked below the line.

The Nernst equation

SLIDES 14 TO 16 OF THE DECK

The Nernst equation gives the equilibrium potential. You will learn to read it, not calculate with it. It turns a concentration ratio into the voltage that would exactly cancel it. At body temperature, 37°C (98.6°F), it can be written $E_{ion} = (61.5 \text{ mV} \div z) \times \log_{10} ([ion] \text{ outside} \div [ion] \text{ inside})$, where z is the charge on the ion: $+1$ for potassium and sodium, -1 for chloride, $+2$ for calcium. The 61.5 mV bundles the gas constant, the Faraday constant, the conversion to log base 10 and body temperature, and many books round it to 61 or 60 . The full form is $E = (RT \div zF) \times \ln(\text{out} \div \text{in})$, which is the same equation before the constants are multiplied out.

Three things are worth reading out of it. Only the ratio of outside to inside matters, so the steeper the gradient, the larger the voltage needed to balance it. Every tenfold difference is worth about 61 mV for an ion with a single charge. The sign comes from two things together, the direction of the gradient and the charge on the ion. A positive ion that is more concentrated inside needs a negative interior to hold it in, so its equilibrium potential is negative, and a positive ion that is more concentrated outside needs a positive interior to hold it out, so its equilibrium potential is positive. A negative ion flips the sign. And an ion with a double charge feels the voltage twice as strongly, so it needs only half the voltage to balance the same ratio, which is why z is in the denominator.

For the concentrations in these notes, the values to know are E_K about -95 mV (many books say -90), E_{Na} about $+61$ mV, E_{Ca} about $+120$ mV and E_{Cl} about -64 mV, close to the resting potential. The most useful thing an equilibrium potential tells you is direction. When an ion's channels open, the ion moves so that it pulls the membrane toward its own equilibrium potential, and the farther the membrane is from that value, the larger the driving force.

$$E_{ion} = (61.5 \text{ mV} \div z) \times \log_{10} ([ion] \text{ outside} \div [ion] \text{ inside})$$

WORKED PROBLEM 1

Potassium is 4 mM outside the cell and 140 mM inside. Without calculating anything, predict whether E_K is positive or negative and explain why. Then say which way potassium moves when its channels are open in a neuron resting at -70 mV, and which way that moves the membrane potential.

Use the two forces, not arithmetic. Potassium carries one positive charge, so $z = +1$.

THE WORK

- Start with the chemical force. Potassium is 35 times more concentrated inside, so its concentration gradient pushes it **out**.
- Ask what electrical force would balance that push. Potassium is a positive ion, so the force that holds it back has to pull it in, and a positive ion is pulled in by a **negative** interior. That is why E_K has to be negative.
- The equation says the same thing in its own way. Outside over inside is 4 over 140 , a ratio smaller than 1 , and for an ion with a positive charge that always gives a negative answer. For these concentrations E_K is about **-95 mV**. Books that use slightly different concentrations say -90 mV, and either value is fine.
- Read that number as a sentence: if potassium were the only ion that could cross, the inside would settle about 95 mV negative, and at that voltage potassium would stop moving on net.
- Now the direction. The neuron is at -70 mV, which is not negative enough to hold potassium in. The outward push from concentration is still larger than the inward pull from voltage, so potassium **leaks out**, slowly and continuously. Each potassium ion that leaves carries a positive charge out, so the membrane moves **down, toward -95 mV**.
- The rule to keep: an ion whose channels are open always moves the membrane toward its own equilibrium potential. The gap between the membrane potential and E_K , here about 25 mV, is potassium's driving force, and it is small.

WORKED PROBLEM 2

Sodium is 145 mM outside and 15 mM inside. Chloride is 110 mM outside and 10 mM inside. Predict the sign of each equilibrium potential from the direction of the gradient and the charge on the ion. Then, using E_{Na} of about +61 mV and E_{Cl} of about -64 mV, say which way each ion moves when its channels open in a neuron resting at -70 mV, and rank potassium, sodium and chloride by the size of their driving force.

Both gradients point inward. Watch the charge on chloride.

THE WORK

- 1** Sodium. It is about 10 times more concentrated outside, so the chemical force pushes it **in**. To hold back a positive ion that is being pushed in, the inside has to repel it, which means the inside has to be positive. E_{Na} is **positive**, about +61 mV.
- 2** Chloride. It is also more concentrated outside, so the chemical force pushes it in as well. But chloride carries a negative charge, and a negative ion is pushed back out by a **negative** interior. So E_{Cl} is **negative**, about -64 mV, even though its gradient points the same way as sodium's. The charge flips the sign, which is what $z = -1$ does in the equation.
- 3** Direction for sodium. The cell is at -70 mV and sodium's equilibrium potential is +61 mV. Concentration and voltage both point inward, so when sodium channels open, sodium rushes **in** and the membrane moves **up** toward +61 mV. Very little gets in at rest only because very few sodium channels are open.
- 4** Direction for chloride. A cell at -70 mV is only about 6 mV more negative than E_{Cl} , so chloride is nearly balanced. Opening chloride channels moves very little of it, and the small net movement, outward at this voltage, nudges the membrane toward -64 mV.
- 5** Rank the driving forces by the gap between -70 mV and each equilibrium potential: sodium about 131 mV, potassium about 25 mV, chloride about 6 mV. Sodium has by far the largest driving force, and at rest it is held back only by closed channels.
- 6** What the three say together: at rest, potassium is slightly off balance, chloride is close to balanced, and sodium is far off balance. That is why opening sodium channels, the subject of the next lecture, changes the voltage so quickly.

and opposite charge, and the pump returns the ions that leaked. This is a steady state and not an equilibrium. An equilibrium needs no energy to stay where it is. A steady state stays constant only while energy keeps being supplied, and here that energy is ATP.

The Goldman equation, also called the Goldman-Hodgkin-Katz or GHK equation, handles several ions at once. As with the Nernst equation, you will learn to read it, not calculate with it. In words, it says that the membrane potential is a blend of the ions' equilibrium potentials, and each ion's share is weighted by the membrane's permeability to that ion, P , which in practice means how many of that ion's channels are open. Written out for potassium and sodium it reads $V_m = 61.5 \times \log_{10} \left(\frac{P_K [K^+]_{out} + P_{Na} [Na^+]_{out}}{P_K [K^+]_{in} + P_{Na} [Na^+]_{in}} \right)$. Each ion's concentrations are multiplied by its permeability, so an ion with few open channels has little say. Chloride can be added as a third term. It is left out here because chloride is already close to balanced at rest.

The pump therefore contributes in two ways. Its direct contribution, from moving 3 charges out for every 2 in, is only a few millivolts. Its indirect contribution is everything else, because without the gradients the leak channels have nothing to work with. The resting potential is produced mainly by potassium leaking out through leak channels, and the pump makes that possible by keeping the cell full of potassium. If a poison blocked every pump, the voltmeter would lose only a few millivolts in the first second, because the gradients are still there. Over the following hours sodium would leak in and potassium out with nothing to return them, both equilibrium potentials would drift toward zero, the membrane potential would follow, and the cell would swell as water followed the sodium in.

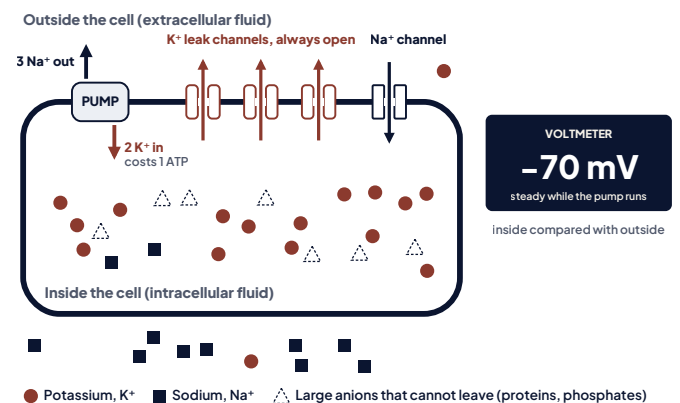


Figure 6. Many potassium leak channels, a few sodium leak channels, and the pump putting both ions back.

The real resting cell, and the Goldman equation

SLIDES 17 TO 19 OF THE DECK

A real membrane has many potassium leak channels and a few sodium leak channels. The small inward sodium leak brings positive charge in and raises the potential from -95 mV to about -70 mV. At -70 mV neither ion is at its own equilibrium, so potassium keeps leaking out and sodium keeps leaking in, all the time. The voltage holds steady because the two leaks carry equal

WORKED PROBLEM 3

At rest the membrane is about 25 times more permeable to potassium than to sodium. Without calculating, predict whether V_m sits nearer E_K (about -95 mV) or E_{Na} (about $+61$ mV), and roughly where. Then predict V_m if P_{Na} fell to zero, and if P_{Na} became very large.

E_K about -95 mV and E_{Na} about $+61$ mV. P_K is about 25 times P_{Na} .

THE WORK

- 1 Read the equation as a tug of war. Each ion pulls the membrane toward its own equilibrium potential, and its pull is weighted by its permeability, P , which in practice means how many of its channels are open.
- 2 Potassium's weight is about 25 times sodium's, so the membrane sits **close to E_K** and nowhere near halfway between the two. Halfway would be about -17 mV, which is nothing like a resting cell.
- 3 Sodium's small weight still counts. Sodium is far from its own equilibrium potential and there is a lot of it outside, so the few open sodium channels let in enough positive charge to lift the potential about 24 mV above E_K . That puts rest at about **-70 mV**. The equation gives about -71 mV for these values, which matches what an electrode measures in a real neuron.
- 4 Check the first extreme. If P_{Na} fell to zero, potassium would be the only ion with any weight, the equation would become the Nernst equation for potassium, and V_m would sit at **-95 mV**. That was cell 1.
- 5 Check the other extreme. If P_{Na} became very large, sodium's weight would swamp potassium's and V_m would head for **$+61$ mV**. That was cell 2, and the next lecture shows a neuron doing something close to it for a thousandth of a second.
- 6 The sentence to keep: the resting potential is a blend of the equilibrium potentials, weighted by permeability, and it sits near E_K because the resting membrane is far more permeable to potassium than to anything else.

Steady state

A condition that stays constant only because energy is continually spent to hold it there. The resting potential is a steady state.

Equilibrium

A condition that stays constant with no energy input. A single ion at its Nernst potential is at equilibrium.

Permeability (P)

How easily an ion crosses the membrane, set mainly by how many of its channels are open.

Goldman (GHK) equation

Predicts the membrane potential from the concentrations of several ions, each weighted by its permeability.

Channels, and the words for a potential that moves

SLIDES 21 TO 22 OF THE DECK

Every ion channel is a protein pore, and channels are sorted by what makes the pore open. Leak channels have no gate and are always open. They are in every cell, and potassium leak channels set the resting potential. Voltage gated channels open when the membrane potential changes. Sodium and potassium versions run along axons, and calcium versions sit at axon terminals and in heart muscle. Ligand gated channels, also called chemically gated, open when a specific molecule binds to them. The acetylcholine receptor on skeletal muscle at the neuromuscular junction is one, and so are the receptors on a neuron's dendrites. Mechanically gated channels open when the membrane is stretched, pressed or bent, as in touch receptors in the skin and hair cells in the inner ear.

Only leak channels are open in a resting cell, which is why this lecture needed only them. A cell uses the other three to change its membrane potential. The Goldman equation tells you what opening any of them does: it raises the permeability for whichever ion fits through, which gives that ion a bigger weight, so the membrane potential moves toward that ion's equilibrium potential. The concentrations do not need to change at all.

Four words describe a membrane potential that moves. A resting membrane is polarized, meaning it has a negative side and a positive side. Depolarization is any change that makes the inside less negative than rest. Overshoot is the part of a depolarization that goes past 0 mV, where the inside is briefly positive, so define it by the zero line and not by the height of the peak. Repolarization is the return toward the resting potential after a depolarization. Hyperpolarization is any change that makes the inside more negative than rest. A hyperpolarized cell is farther from threshold and harder to excite. A depolarized cell is closer to threshold and, at least at first, easier to excite.

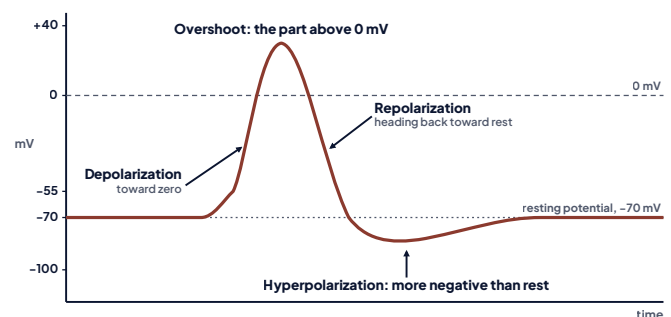


Figure 7. Each word describes a direction of change compared with the resting potential.

Voltage gated channel

Opens in response to a change in membrane potential. Found along axons, at axon terminals and in muscle.

Ligand gated channel

Opens when a specific molecule, such as a neurotransmitter, binds to it.

Mechanically gated channel

Opens when the membrane is stretched, pressed or bent.

Threshold

The membrane potential, about -55 mV in a neuron, at which voltage gated sodium channels open and the cell fires. Covered in the next lecture.

Changing the blood

SLIDES 23 TO 24 OF THE DECK

Plasma potassium is the outside potassium for every cell in the body, so a change in the blood is a change in the Nernst ratio for every cell at once. More potassium outside means a weaker gradient pushing potassium out, less negative charge left behind, and every cell depolarizes. Less potassium outside means a steeper gradient, and every cell hyperpolarizes.

With low potassium, rest moves away from threshold. Nerve and skeletal muscle cells become harder to excite, and the patient has muscle weakness and cramps. Heart muscle responds to low potassium in a more complicated way, which we take up in the cardiac weeks. With high potassium, rest moves toward threshold, and for a short time cells fire more easily. Then a second effect becomes more important. Voltage gated sodium channels that stay at a depolarized potential for too long switch to an inactivated state and cannot reopen until the membrane repolarizes. Cells become unable to fire. In skeletal muscle that is weakness. In the heart it is slowed, disorganized conduction.

Now compare sodium. Dropping plasma sodium by 20 mM, from 145 to 125 , moves E_{Na} from $+61$ to $+57$ mV and the resting potential from -71 to -73 mV, a shift of 2 mV. Raising potassium by only 3 mM moved the resting potential 7 mV. There are two reasons, and both are in the equations. The resting membrane is about 25 times more permeable to potassium, so the resting potential follows E_K and mostly ignores E_{Na} . And the Nernst equation works on ratios: 4 to 7 mM nearly doubles the outside potassium, while 145 to 125 mM changes sodium by only 14 percent. Low sodium is still dangerous, but for a different reason. It lowers the osmolarity of the extracellular fluid, water moves into brain cells, and they swell.

Same cell, three different blood potassium values

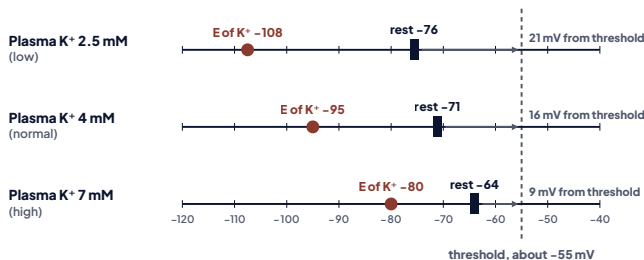


Figure 8. The same cell at three blood potassium values. Threshold is covered fully in the next lecture.

WORKED PROBLEM 4

Predict which way E_K and the resting potential move when plasma potassium rises from 4 to 7 mM, and when it falls to 2.5 mM. For each, say whether the cell moves toward threshold or away from it, and why.

Inside potassium stays at 140 mM, and nothing else changes.

THE WORK

- 1 High, 7 mM.** More potassium outside makes the gradient less steep, so less negative charge inside is needed to balance it. E_K becomes **less negative**, moving from about -95 toward about -80 mV. The resting potential follows E_K because potassium has most of the permeability, so the cell **depolarizes**, from about -71 to about -64 mV, and moves closer to threshold.
- 2 Low, 2.5 mM.** Less potassium outside makes the gradient steeper, so a more negative interior is needed to hold potassium in. E_K becomes **more negative**, about -108 mV, and the resting potential follows it down to about -76 mV. The cell **hyperpolarizes** and moves farther from threshold.
- 3** Why the direction surprises people: adding a positive ion outside makes the inside less negative, not more. The reason is not the charge on the added potassium. A weaker push out means less potassium leaves, so less negative charge is left behind inside.
- 4** With low potassium, rest moves away from threshold. Nerve and skeletal muscle cells become harder to excite, and the patient has muscle weakness and cramps. Heart muscle responds to low potassium in a more complicated way, which we take up in the cardiac weeks.
- 5** With high potassium, rest moves toward threshold, and for a short time cells fire more easily. Then a second effect becomes more important. Voltage gated sodium channels that stay at a depolarized potential for too long switch to an inactivated state and cannot reopen until the membrane repolarizes. Cells become unable to fire. In skeletal muscle that is weakness. In the heart it is slowed, disorganized conduction.

The patient, explained

SLIDE 25 OF THE DECK

Most of the potassium in the body is inside skeletal muscle cells, at 140 mM. Five hours of crushing killed a large mass of muscle, and dead cells release their contents. While the worker was pinned, the wall acted like a tourniquet. No blood flowed through his legs, so the released potassium stayed there. When the wall was lifted, blood flow returned and carried that potassium into his circulation all at once, and plasma potassium rose from 4 to 7 mM.

From there it is worked problem 4. E_K moved from -95 to -80 mV in every cell of his body, and resting potentials followed it upward by about 7 mV. Heart cells held at a depolarized potential have many of their sodium channels held in the inactivated state. The electrical signal crosses the heart slowly and unevenly, the rhythm breaks down, and the heart stops pumping. That is why a 3 mM change in potassium affects the heart when a 3 mM change in sodium has almost no effect.

Look at the treatment as an experiment. The variable that failed is plasma potassium, and through it the gap between rest and threshold. Calcium given into a vein does not change potassium at all. Raising the calcium outside the cell moves threshold to a less negative value, which restores the gap, and it works within minutes. Insulin with glucose, and albuterol, both speed up the sodium potassium pump, so potassium moves out of the plasma and into cells within the hour, although it is still in the body. Dialysis, or working kidneys, is the only step that removes potassium from the body. The same physiology shows up whenever many cells break open at once, as in large burns, and whenever the kidneys cannot excrete potassium, as in kidney failure.

What this lecture assumes you already have

If any line here is shaky, fix it before you work the lecture. The review pages are linked from each card.

CHEMISTRY

What an ion is, and the charges on Na^+ , K^+ , Cl^- and Ca^{2+}
Opposite charges attract and like charges repel
Diffusion: particles move from higher concentration toward lower
Concentration units: moles per liter and millimolar (mM)

[Chemistry review page](#)

FROM EARLIER WEEKS

The lipid bilayer, and why an ion cannot cross its oily middle unaided (Week 3)
Intracellular and extracellular fluid, and that a blood test reads the extracellular side (Week 3)
Channels and pumps as membrane proteins, and that ATP pays for uphill transport

No new anatomy is needed for this lecture, so there is no anatomy list this time.

MATH

Reading a voltage scale with negative values: -90 mV is more negative, and lower on the scale, than -70 mV
The gap between two points on that scale, for example -70 mV to $+60$ mV is 130 mV
What it means when a ratio such as outside over inside is bigger than 1 or smaller than 1

Competency 6, the membrane potential simulation, is done in lab. Pick one prompt per competency. Blank paper, nothing open.

Read the full instructions first, including how the first pass before the lecture differs from the second pass after it.

01

Ion distribution and electrochemical gradients

State the typical intracellular and extracellular concentrations of sodium, potassium, chloride, and calcium and separate the chemical from the electrical component of the driving force on each ion.

Do one of the two. Head your paper with the number.

1-1

Draw a cell and give each ion a pair of bars, inside and outside, height for concentration: sodium 145 mM out against 15 mM in, potassium 4 out against 140 in, chloride 110 out against 10 in, calcium about 1 out against 0.0001 free in. Mark the membrane inside negative. On each ion draw two separate arrows, chemical force and electrical force. One line: two ions have arrows pointing opposite ways, so name both and say which force wins.

1-2

Draw four vertical voltage scales side by side, one per ion, from minus 100 mV up to plus 150 mV. Mark each ion's equilibrium potential: potassium near minus 90, sodium near plus 60, calcium near plus 120, chloride near minus 65. Draw one horizontal line at minus 70 mV across all four. One line: each gap from that line is a driving force, so name the ion with the largest gap and say what stops it moving at rest.

Close the page and produce it

These are the Week 4 competencies this lecture covers, straight off the study guide, with the two prompts that go with each one.

02**Nernst equation**

Explain what an ion's equilibrium potential means, predict its sign from the direction of the ion's concentration gradient and its charge, and use it to predict which way that ion will move when its channels open.

Do one of the two. Head your paper with the number.

2-1

Draw a cell with potassium high inside and sodium high outside. For each ion, draw the chemical arrow (down its concentration gradient) and the electrical arrow that would balance it. Beside each ion, write its equilibrium potential and whether the sign is positive or negative, and one line on why the sign comes out that way.

2-2

Draw a vertical voltage scale from minus 100 to plus 70 mV. Mark E_K , E_{Na} and a resting potential of minus 70 mV. Now open potassium channels, then sodium channels: for each, draw an arrow showing which way the ion moves and which way the membrane potential moves, and label which one has the bigger driving force and why.

03**Resting membrane potential**

Explain why the resting membrane potential sits near the potassium equilibrium potential, explain in words how the Goldman equation combines each ion's concentration gradient and permeability, and predict how the potential shifts when membrane permeability to potassium or sodium changes.

Do one of the two. Head your paper with the number.

3-1

Draw a cell at rest with minus 70 mV written inside. On its membrane draw potassium leak channels as many, sodium leak channels as few, and the sodium potassium pump. Beside the cell draw a voltage scale with the potassium equilibrium potential near minus 90 and the sodium equilibrium potential near plus 60, and mark where minus 70 sits between them. One line: the resting potential sits close to one of those two and nowhere near halfway, so say what makes it lean.

3-2

Draw a voltage scale from minus 90 to plus 60 mV with the resting potential marked, then draw four labeled arrows off it showing where the potential moves when you open more potassium channels, open sodium channels, block the potassium leak, and stop the sodium potassium pump. One line: one of those four shifts the potential only a few mV even though it eventually ruins the cell, so name it and say why the immediate effect is so small.

04

Ion channel gating

Compare leak, voltage gated, ligand gated, and mechanically gated channels by what opens each one and give a physiological location for each.

Do one of the two. Head your paper with the number.

4-1

Draw four channels in a row in a strip of membrane: leak, voltage gated, ligand gated, and mechanically gated. Draw each one twice, closed and open, and draw whatever opens it right beside it, a voltage change, a bound molecule, a stretch or a bend. Under each write one real place in the body it is found. One line: only one of the four is open at rest, so say what job that gives it.

4-2

Draw one neuron stretched across the whole width of your box. Along it place each channel type where it actually works: ligand gated on the dendrites, voltage gated sodium and potassium along the axon, leak channels everywhere, voltage gated calcium at the terminal. Add a mechanically gated channel on a sensory ending drawn off to the side. One line: number the order these open during one real signal and say which one starts it.

05

Depolarization and hyperpolarization

Define depolarization, repolarization, hyperpolarization, and overshoot and label each on a membrane potential tracing.

Do one of the two. Head your paper with the number.

5-1

Draw a membrane potential tracing with time on the x and mV on the y, starting at minus 70. Take it up past zero to about plus 30 and back down below minus 70 before it returns. Bracket and label depolarization, overshoot, repolarization, hyperpolarization, and the return to rest. Draw zero mV as a dashed line. One line: define overshoot using that dashed line, not using the peak.

5-2

Draw one set of axes with a dashed line at minus 70 mV and another at zero. Plot four traces from rest: a small depolarization that fades away, a hyperpolarization, a depolarization that crosses threshold, and a repolarization from plus 30 back down. Label each with its correct term. One line: say which of your traces make the cell easier to fire and which make it harder, and how you can tell from direction alone.

Dr. Sharilyn Rennie · BIO 005 Human Physiology · The resting membrane potential. Concentrations are typical teaching values, not laboratory reference ranges. All figures are original drawings for this course.